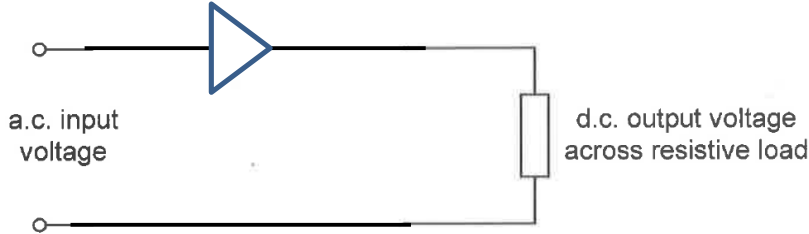
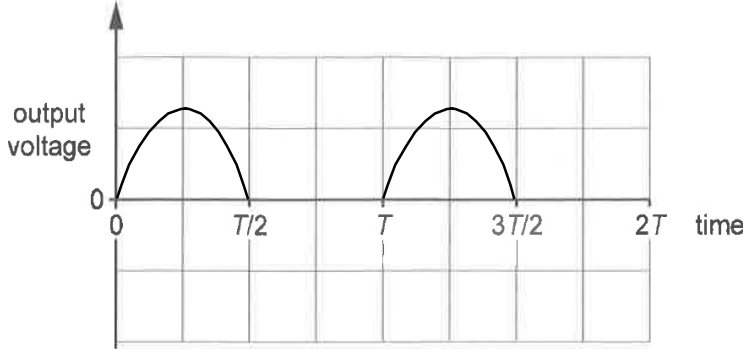


|  |                                                                                                                                                                                                        |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | 7(a)(i)                                                                                                                                                                                                |
|  | $\frac{\text{cost to travel 1.0 km in the typical EV}}{\text{cost to travel 1.0 km in an ICE car}} = \frac{\left( \frac{0.23 \times 72}{400} \right)}{\left( \frac{80}{800} \right)} = 0.414$          |
|  | 7(a)(ii)                                                                                                                                                                                               |
|  | <p>In Singapore, a car typically travels less than 40 km a day or 280 km a week while the range of a EV is 400 km. As such, EVs in Singapore typically need to be charged less than once per week.</p> |

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7(b)(i)   | <p>Typical home battery charger provides 7.2 kW at 32 A.</p> <p>Since <math>P = I V</math>, voltage supply <math>V = \left( \frac{7200}{32} \right) = 225 \text{ V}</math>.</p> <p>As each cell can only takes 2.0 A, the max current supply of 32 A is supplied in parallel circuit of <math>\left( \frac{32}{2} \right) = 16</math> branches.</p> <p>Max voltage across each cell is 3.0 V, the voltage supply of 225 V is across <math>\left( \frac{225}{3} \right) = 75</math> cells in series.</p> <p>Minimum number of cells = <math>16 \times 75 = 1200</math> cells.</p> |
| 7(b)(ii)  | <p>Since when fully charged, the range is 400 km and it takes 10 hours to fully charge the battery, 'charging speed' of <math>40 \text{ km h}^{-1}</math> refers to the distance that can be covered by one hour of charging.</p>                                                                                                                                                                                                                                                                                                                                                |
| 7(b)(iii) | <p>It is the amount of energy available per kg of the battery.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 7(b)(iv)  | <p>Battery capacity is 72 kWh.</p> <p>mass = <math>\frac{72000}{141} = 510.6 \text{ kg} = 510 \text{ kg}</math> (2 s.f.)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 7(b)(v)   | <p>The battery is placed under the floor of the EV to lower the C.G. of the car for added stability.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 7(c)(i)   | <p>Add a single diode in series.</p>  <p style="text-align: center;"><b>Fig. 7.1</b></p>                                                                                                                                                                                                                                                                                                                                                                                                     |
| 7(c)(ii)  | <p>Single diode half wave rectified waveform.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|          |                                                                                                                                                                                                                                                                                                                                               |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7(d)     | <p>1 kWh = <math>1000 \times 3600 = 3.6 \times 10^6</math> J<br/> 72kWh gives a range of 400 km.</p> <p>K.E. used to charge the battery = <math>\frac{1}{2}mv^2 = \frac{1}{2}(1685)(25)^2 = 5.26 \times 10^5</math> J</p> <p>Distance the EV can travel = <math>\frac{400}{72 \times 3.6 \times 10^6} (5.26 \times 10^5) = 0.81</math> km</p> |
| 7(e)(i)  | <p>A wireless charger works based on electromagnetic induction where a changing magnetic flux linkage generated by the a.c. voltage flowing through a coil will generate an induced e.m.f. for charging.</p>                                                                                                                                  |
| 7(e)(ii) | <p>This is to prevent the loss of magnetic flux as it's strength will decrease with distance between the charging pad and the battery.</p>                                                                                                                                                                                                    |
| 7(f)(i)  | <p>Maximum output torque is 395 Nm.</p> <p>Torque of the couple = <math>NF d = N(BIL)d = NBI(Ld) = NBIA</math></p> <p><math>395 = (1200)B(96)(6.1 \times 10^{-3})</math></p> <p><math>B = 0.56</math> T</p>                                                                                                                                   |
| 7(f)(ii) | <p>With four pairs of magnetic poles, the magnetic force acting on the coil will be at its maximum values on more parts of the rotation and this will result in a smoother turning of the coil and thus smoother running of the motor.</p>                                                                                                    |